

CHE 101 POSSIBLE QUESTION

PLEASE NOTE: This concept is brought to you to assist your read skills and test your ability in preparation for test and examination



- Reactions in which reactants react to form products and simultaneously products reverse back in to reactants are called
 - Reversible reactions
 - Irreversible reaction
 - Spontaneous reactions
 - non spontaneous reactions

Answer: A

- Conditions for equilibrium of concentration of reactant and products will be same if
 - temperature is constant
 - pressure or volume is constant
 - concentration of products and reactants are same
 - all of above

Answer: D

- Equilibrium constant can be used to
 - predict direction of chemical reaction
 - predict extent of chemical reaction
 - determine the equilibrium concentration of mixture
 - all of above

Answer: D

- Rate at which reaction proceeds is directly related to
 - active masses of reactants

- sum of active masses of reactants
- product of active masses of reactants
- ratio of active masses of reactants

Answer: C

- Smallest particle of an element which can take part in any chemical change.
 - nucleus
 - atom
 - proton
 - neutron

Answer: B

- Number of times a p^+ is heavier than an e^- is
 - 18 times
 - 184 times
 - 200 times
 - 1840 times

Answer: D

- Chemical elements are represented by Greek or Latin
 - alphabets
 - symbols
 - gods
 - illusions

Answer: B

- Atomic number is number of
 - protons (p^+)
 - electrons (e^-)
 - neutrons (n^0)
 - nucleons (p^+ and n^0)

Answer: A

- An Electric field deflects beams of
 - protons
 - electrons
 - neutrons
 - both A and B

Answer: D

- Modern method for separation of isotopes is
 - laser separation
 - chromatography
 - ionization
 - Xray

Answer: A

- In new spectrometers each ion hits a
 - Detector

- B. ionizer
- C. collector
- D. graph

Answer: A

12. Mass spectrometer use to determine isotopes in solid state is

- A. Bohr's
- B. Aston's
- C. dempester's
- D. Alison's

Answer: C

13. In accelerating chamber of mass spectrometer potential difference is

- A. 500-2000
- B. 600-7000
- C. 300-8000
- D. 700-9000

Answer: A

14. Instrument use to collect ions is

- A. Electrometer
- B. ionizer
- C. spectrometer
- D. none

Answer: A

1. Electron was discovered by_____.

- 1. Chadwick
- 2. Thomson
- 3. Goldstein
- 4. Bohr

Answer: 2

2. An atom has a mass number of 23 and atomic number 11. The number of protons are_____.

- 1. 11
- 2. 12
- 3. 23
- 4. 44

Answer: 1

3. The mass of the atom is determined by_____.

- 1. neutrons
- 2. neutron and proton
- 3. electron
- 4. electron and neutron

Answer: 2

4. The K, L and M shells of an atom are full. Its atomic number is_____.

- 1. 18
- 2. 20
- 3. 10
- 4. 12

Answer: 1

5. Which of the air pressures is appropriate for the production of cathode rays in the discharge tube?

- 1. 1 cm Hg
- 2. 1 mm Hg
- 3. 0.001 cm Hg
- 4. 0.001 mm Hg

Answer: 4

6. Cathode rays are deflected towards_____.

- 1. positive electrode
- 2. negative electrode
- 3. both electrodes
- 4. none of the electrodes

Answer: 1

7. The absolute charge of an electron is_____.

- 1. $-1.6 \times 10^{-19}C$
- 2. $+1.6 \times 10^{-19}C$
- 3. $1.6 \times 10^{-19}C$
- 4. $16 \times 10^{-19}C$

Answer: 1

8. The proton is heavier than an electron by_____.

1. 1850 times
2. 1840 times
3. 1000 times
4. 100 times

Answer: 2

9. Carbon-12 atom has_____.

1. 6 electrons, 6 protons, 6 neutrons
2. 6 electrons, 12 protons, 6 neutrons
3. 12 electrons, 6 protons, 6 neutrons
4. 18 electrons, 6 protons and 6 neutrons

Answer: 1

10. Chadwick got the Nobel Prize for the discovery of _____.

1. protons
2. neutrons
3. electrons
4. mesons

Answer: 2

11. Mass number is equal to the_____.

1. number of protons + number of electrons
2. number of protons + number of neutrons
3. number of neutrons + number of electrons
4. number of electrons

Answer: 2

12. The element X has 2 valence electrons. It is a_____.

1. metal
2. non-metal
3. metalloid
4. gas

Answer: 1

13. The volume of the nucleus of an atom when compared to the extra nuclear part is_____.

1. bigger
2. smaller
3. same size
4. unpredictable

Answer: 2

14. In Rutherford's alpha-scattering experiment, a foil of element that was used _____.

1. gold
2. silver
3. aluminum
4. magnesium

Answer: 1

15. An element has an electronic configuration of 2, 8, 7. Its valency is

1. 1
2. 7
3. 17
4. 8

Answer: 1

16. The other name of ${}^1_1\text{H}$ is _____

1. protium
2. tritium
3. deuterium
4. proton

Answer: 1

17. During a chemical reaction, atomic number_____.

1. changes
2. remains same
3. changes and then is restored
4. changes alternately

Answer: 2

18. The fixed circular paths around the nucleus are called_____.

1. orbits
2. orbitals

1. The formula weight of the compound, $\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$ is:

- (a) 394.4 g (b) 666.4 g (c) 110,900 g (d) 466.8 g (e) 561.2 g

2. The weight of a millimole of $(\text{NH}_4)_2\text{HPO}_4$ is:

3. nucleons
4. mesons

Answer: 1

- (a) 132 g (b) 114 g (c) 1.14×10^{-3} g (d) 0.132 g (e) 6.02×10^{20} g

3. How many moles of alanine, $\text{C}_3\text{H}_7\text{NO}_2$, are there in 159 g of alanine?

- (a) 1.42×10^4 (b) 1.78 (c) 0.992 (d) 0.560 (e) 3.31

Answer: 1. (b) 2. (d) 3. (b)

Sir TOSIN

1. Which of the following has a positive charge? (a) Proton (b) neutron (c) anion (d) electron (e) atom

2. Rutherford carried out experiments in which a beam of alpha particles was directed at a thin piece of metal foil. From these experiments he concluded that: (a) electrons are massive particles. (b) The positively charged parts of atoms are moving about with a velocity approaching the speed of light. (c) The positively charged parts of atoms are extremely small and extremely heavy particles. (d) The diameter of an electron is approximately equal to that of the nucleus. (e) Electrons travel in circular orbits around the nucleus.

3. Consider the species ^{72}Zn , ^{75}As and ^{74}Ge . These species have:

- (a) the same number of electrons.
- (b) the same number of protons.
- (c) the same number of neutrons.
- (d) the same number of protons and neutrons.
- (e) the same mass number.

4. The neutral atoms of all of the isotopes of the same element have

- (a) different numbers of protons.
- (b) equal numbers of neutrons.
- (c) the same number of electrons.

- (d) the same mass numbers.
- (e) the same masses.

5. What is the atomic weight of a hypothetical element consisting of two isotopes, one with mass = 64.23 amu (26.0%), and one with mass = 65.32 amu?

- (a) 65.3 amu
- (b) 64.4 amu
- (c) 64.9 amu
- (d) 65.0 amu
- (e) 64.8 amu

6. Naturally occurring rubidium consists of just two isotopes. One of the isotopes consists of atoms having a mass of 84.912 amu; the other of 86.901 amu. What is the percent natural abundance of the heavier isotope?

- (a) 15%
- (b) 28%
- (c) 37%
- (d) 72%
- (e) 85%

7. What is the frequency of light having a wavelength of 4.50×10^{-6} cm?

- (a) $2.84 \times 10^{-12} \text{ s}^{-1}$
- (b) $2.10 \times 10^4 \text{ s}^{-1}$
- (c) $4.29 \times 10^{14} \text{ s}^{-1}$
- (d) $1.06 \times 10^{22} \text{ s}^{-1}$
- (e) $6.67 \times 10^{15} \text{ s}^{-1}$

8. The emission spectrum of gold shows a line of wavelength 2.676×10^{-7} m. How much energy is emitted as the excited electron falls to the lower energy level?

- (a) 7.43×10^{-19} J
- (b) 5.30×10^{-20} J
- (c) 6.05×10^{-19} J
- (d) 3.60×10^{-20} J
- (e) 5.16×10^{-20} J

9. Which of the responses contains all the statements that are consistent with the Bohr theory of the atom (and no others)?

- (1) An electron can remain in a particular orbit as long as it continually absorbs radiation of a definite frequency.
- (2) The lowest energy orbits are those closest to the nucleus.
- (3) An electron can jump from the K shell ($n = 1$ major energy level) to the M shell ($n = 3$ major energy level) by emitting radiation of a definite frequency.
- (a) 1, 2, 3 (b) 2 only (c) 3 only (d) 1, 2 (e) 2, 3

10. The Heisenberg Principle states that _____.

- (a) No two electrons in the same atom can have the same set of four quantum numbers.
- (b) Two atoms of the same element must have the same number of protons.
- (c) It is impossible to determine accurately both the position and momentum of an electron simultaneously.
- (d) Electrons of atoms in their ground states enter energetically equivalent sets of orbitals singly before they pair up in any orbital of the set.
- (e) Charged atoms (ions) must generate a magnetic field when they are in motion.

11. Which statement about the four quantum numbers which describe electrons in atoms is **incorrect**?

- (a) n = principal quantum number, $n = 1, 2, 3, \dots$
- (b) l = subsidiary (or azimuthal) quantum number, $l = 1, 2, 3, \dots, (n+1)$

(c) m_l = magnetic quantum number, $m_l = (-l), \dots, 0, \dots, (+l)$

(d) m_s = spin quantum number, $m_s = +1/2$ or $-1/2$.

(e) The magnetic quantum number is related to the orientation of atomic orbitals in space.

12. Which atomic orbital is spherical in shape? (Note: you should know and be able to recognize the shapes of the s orbital, p_x , p_y , and p_z orbitals, and d_{xy} , d_{yz} , d_{xz} , $d_{x^2-y^2}$ and d_{z^2} orbitals.)

- (a) 2s (b) 3p (c) 3d (d) 4f
- (e) They are all spherical

13. The maximum number of electrons that can be accommodated in a sublevel for which $l = 3$ is:

- (a) 2
- (b) 10
- (c) 6
- (d) 14
- (e) 8

14. The ground state electron configuration for arsenic is:

- (a) $[\text{Ar}] 4s^2 4p^{13}$
- (b) $[\text{Kr}] 4s^2 4p^1$
- (c) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{12} 4s^2 4p^1$
- (d) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8 4p^5$
- (e) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^3$

15. Which of the following electron configurations is correct for nickel?

- (a) $[\text{Ar}] 4s^1 3d^8$
- (b) $[\text{Kr}] 4s^1 4d^8$
- (c) $[\text{Kr}] 4s^1 3d^8$
- (d) $[\text{Kr}] 4s^2 3d^8$
- (e) $[\text{Ar}] 4s^2 3d^8$

16. The outer electronic configuration $ns^2 np^4$ corresponds to which one of the following elements in its ground state?

- (a) As
- (b) Ca
- (c) Cr
- (d) Br
- (e) S

17. In the ground state of a cobalt atom there are _____ unpaired electrons and the atom is _____.

- (a) 3, paramagnetic
- (b) 5, paramagnetic
- (c) 2, diamagnetic
- (d) 0, diamagnetic
- (e) 2, paramagnetic

18. Which one of the following sets of quantum numbers **could** be those of the distinguishing (last) electron of Mo?

- (a) $n = 4, l = 0, m_l = 0, m_s = +1/2$
- (b) $n = 5, l = 1, m_l = 9, m_s = -1/2$
- (c) $n = 4, l = 2, m_l = -1, m_s = +1/2$
- (d) $n = 5, l = 2, m_l = +2, m_s = -1/2$
- (e) $n = 3, l = 2, m_l = 0, m_s = +1/2$

19. How many p electrons are there in an atom of rubidium?

- (a) 12
- (b) 18
- (c) 24
- (d) 9
- (e) 6

20. A neutral atom of an element has 2 electrons in the first energy level, 8 in the second energy level and 8 in the third energy level. This information does **not** necessarily tell us:

- (a) The atomic number of the element.
- (b) Anything about the element's chemical properties.
- (c) The total number of electrons in s orbitals.
- (d) The total number of electrons in p orbitals.
- (e) The number of neutrons in the nucleus of an atom of the element.

Answers:

Answer: 1. (a) 2. (c) 3. (c) 4. (c) 5. (d) 6. (b) 7. (e) 8. (a) 9. (b) 10. (c) 11. (b) 12. (a) 13. (d) 14. (e) 15. (e) 16. (e) 17. (a) 18. (c) 19. (b) 20. (e)

19. The mass (in grams) of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ required for preparation of 125 mL of 0.90 M solution is:

- (a) 16 g
- (b) 25 g
- (c) 13 g
- (d) 31 g
- (e) 43 g

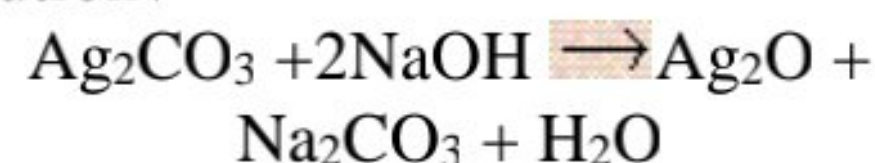
20. What is the molarity of phosphoric acid in a solution labeled 20.0% phosphoric acid (H_3PO_4) by weight with a density = 1.12 g/mL?

- (a) 0.98 M
- (b) 2.3 M
- (c) 2.7 M
- (d) 3.0 M
- (e) 3.6 M

21. How many mL of 17 M NH_3 must be diluted to 500.0 mL to make a 0.75 M solution?

- (a) 13 mL
- (b) 22 mL
- (c) 39 mL
- (d) 73 mL
- (e) None of these

22. How many grams of Ag_2CO_3 are required to react with 28.5 mL of 1.00 M NaOH solution?



- (a) 7.87 g
- (b) 3.93 g
- (c) 15.7 g
- (d) 10.8 g
- (e) 8.16 g

23. How many milliliters of 0.200 M NH_4OH are needed to react with 12.0 mL of 0.550 M FeCl_3 ?



- (a) 99.0 mL
- (b) 33.0 mL
- (c) 8.25 mL
- (d) 68.8 mL
- (e) 132 mL

24. When 250. mL of a 0.15 M solution of ammonium sulfide $(\text{NH}_4)_2\text{S}$ is poured into 120. mL of a 0.053 M solution of cadmium sulfate CdSO_4 , how many grams of a yellow precipitate of cadmium sulfide CdS are formed? The other product is $(\text{NH}_4)_2\text{SO}_4$. (Hint: Write out and balance the equation. Is this a limiting reagent problem?)

- (a) 5.4 g
- (b) 0.92 g
- (c) 2.6 g
- (d) 1.9 g
- (e) 530 g

Answer: 19. (d) 20. (b) 21. (b) 22. (b) 23. (a) 24. (b)

1) Which of the following statements is correct for an electron that has the quantum numbers $n = 4$ and $m_l = -2$?

- A. the electron may be in a p orbital
- B. the electron may be in a d orbital
- C. the electron is in the second principal shell
- D. the electron must have a spin quantum number $m_s = +\frac{1}{2}$

Answer: Since $m_l = -2$, we know that there must be five orientations of atomic orbitals: -2, -1, 0, 1, 2. When the magnetic quantum number, m_l , has five possibilities, the orbitals are d orbitals.

2) The set of quantum numbers, $n = 4$, $l = 3$, $m_l = 2$

- A. describes an electron in a 3p orbital
- B. is not allowed
- C. describes an electron in a 4d orbital
- D. describes one of seven orbitals of a similar type

Answer: When $l = 3$, we are speaking of f orbitals. There are seven f orbitals designated by $m_l = -3, -2, -1, 0, 1, 2, 3$.

1. Quantum numbers are
- A. arithmetical values
 - B. numerical values
 - C. geometric values
 - D. logical values

Answer: B

3) Which of the following statements is correct for an electron that has the quantum numbers $n = 5$ and $m_l = -4$?

- A. the electron may be in a f orbital
- B. the electron must have a spin quantum number $m_s = -1/2$
- C. the electron may be in a p orbital
- D. none of the above

Answer: None of these answers are correct because when $m_l = -4$, we must have at least 9 orbitals, i.e. -4,...0,...,+4.

4) Identify the subshell in which electrons with the quantum numbers $n = 3$, $l = 2$ may be found.

- A. 3d
- B. 4f
- C. 4d
- D. 3p

Answer: Since $n = 3$, we know the main energy level is 3. With $l = 2$, we know that we are speaking of d orbitals. Therefore, 3d seems right.

2. Use of quantum numbers is to specify electron's

- A. energy
- B. position
- C. particles
- D. velocity

Answer: B

3. Azimuthal quantum number actually represents

- A. shells
- B. sub shells
- C. energy
- D. none

Answer: B

4. Quantum numbers are used to describe

- A. proton
- B. nucleus
- C. electron
- D. positron

Answer: C

5. Letter s for azimuthal quantum number stands for

- A. sharp
- B. strong
- C. sufficient
- D. simple

Answer: A

6. In Bohr's model magnetic field which splits spectral lines is

- A. weak
- B. moderate

1. When the system $A + B \rightleftharpoons C + D$ is at equilibrium,

- (a) The sum of the concentrations of A and B must equal the sum of the concentrations of C and D.
- (b) The forward reaction has stopped.
- (c) Both the forward and the reverse reactions have stopped.
- (d) The reverse reaction has stopped.
- (e) Neither the forward nor the reverse reaction has stopped.

C. strong

D. none

Answer: C

7. Value of 'm' depends upon values of

- A. s
- B. l
- C. d
- D. f

Answer: B

8. Sub shell which is not sub-divided into any other orbital

- A. p-sub shell
- B. s-sub shell
- C. d-sub shell
- D. f-sub shell

Answer: B

9. Number of degenerate orbitals in f-sub shell are

- A. six
- B. five
- C. seven
- D. three

Answer: C

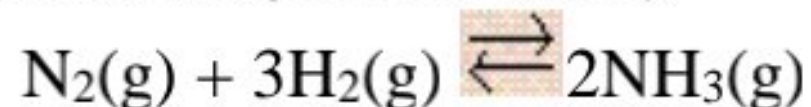
2. $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$

The conventional equilibrium constant expression (K_c) for the system as described by the above equation is:

- (a) $[\text{SO}_2]^2/[\text{SO}_3]$
- (b) $[\text{SO}_2]^2[\text{O}_2]/[\text{SO}_3]^2$
- (c) $[\text{SO}_3]^2/[\text{SO}_3]^2[\text{O}_2]$
- (d) $[\text{SO}_2][\text{O}_2]$
- (e) none of these

3. Consider the following reversible reaction. In a 3.00 liter container, the following amounts are found in equilibrium

at 400 °C: 0.0420 mole N₂, 0.516 mole H₂ and 0.0357 mole NH₃. Evaluate K_c.



- (a) 0.202
- (b) 1.99
- (c) 16.0
- (d) 4.94
- (e) 0.503

4. If the equilibrium constant for the reaction $\text{A} + 2\text{B} \rightleftharpoons \text{C} + 5/2 \text{D}$ has a value of 4.0, what is the value of the equilibrium constant for the reaction $2\text{C} + 5\text{D} \rightleftharpoons 2\text{A} + 4\text{B}$ at the same temperature?

- (a) 0.25
- (b) 0.063
- (c) 2.0
- (d) 8.0
- (e) 16

5. At 445°C, K_c for the following reaction is 0.020. $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$

A mixture of H₂, I₂, and HI in a vessel at 445°C has the following concentrations: [HI] = 2.0 M, [H₂] = 0.50 M and [I₂] = 0.10 M. Which one of the following statements concerning the reaction quotient, Q_c, is **TRUE** for the above system?

- (a) Q_c = K_c; the system is at equilibrium.
- (b) Q_c is less than K_c; more H₂ and I₂ will be produced.
- (c) Q_c is less than K_c; more HI will be produced.
- (d) Q_c is greater than K_c; more H₂ and I₂ will be produced.
- (e) Q_c is greater than K_c; more HI will be produced.

6. Nitrosyl chloride, NOCl, dissociates on heating as shown below. When a 1.50 gram sample of pure NOCl is heated at 350°C in a volume of 1.00 liter, the percent dissociation is found to be 57.2%. Calculate K_c for the reaction as written.



- (a) 0.876

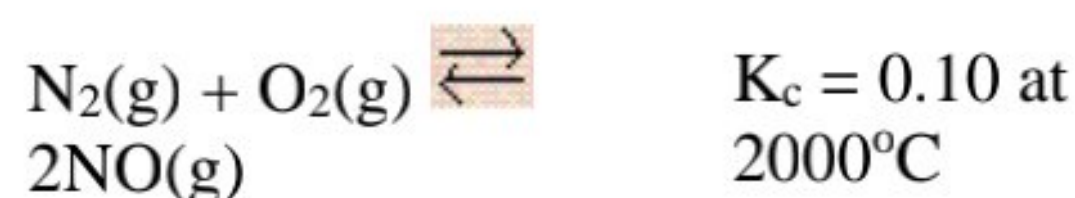
- (b) 9.26
- (c) 0.107
- (d) 1.75×10^{-4}
- (e) 0.0421

7. A quantity of HI was sealed in a tube, heated to 425°C and held at this temperature until equilibrium was reached. The concentration of HI in the tube at equilibrium was found to be 0.0706 mol/L. Calculate the equilibrium concentration of H₂ (and I₂). For the gas-phase reaction,



- (a) $9.55 \times 10^{-3} \text{ M}$
- (b) $1.17 \times 10^{-3} \text{ M}$
- (c) $1.85 \times 10^{-4} \text{ M}$
- (d) $4.78 \times 10^{-3} \text{ M}$
- (e) $2.34 \times 10^{-3} \text{ M}$

8. Consider the reaction:



Starting with initial concentrations of 0.040 mol/L of N₂ and 0.040 mol/L of O₂, calculate the equilibrium concentration of NO in mol/L

- (a) 0.0055 mol/L
- (b) 0.0096 mol/L
- (c) 0.011 mol/L
- (d) 0.080 mol/L
- (e) 0.10 mol/L

9.

K_c = 0.040 for the system below at 450°C. If a reaction is initiated with 0.40 mole of Cl₂ and 0.40 mole of PCl₃ in a 2.0 liter container, what is the equilibrium concentration of Cl₂ in the same system?



- (a) 0.07 M
- (b) 0.16 M
- (c) 0.11 M
- (d) 0.04 M
- (e) 0.26 M

10. The reversible reaction:

$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ has come to equilibrium in a vessel of specific volume at a given temperature. Before the reaction began, the concentrations of the reactants were 0.060 mol/L of SO_2 and 0.050 mol/L of O_2 . After equilibrium is reached, the concentration of SO_3 is 0.040 mol/L. What is the equilibrium concentration of O_2 ?

- (a) 0.010 M
- (b) 0.020 M
- (c) 0.030 M
- (d) 0.040 M
- (e) None of these

11. Consider the gas-phase equilibrium system represented by the equation:

$2\text{H}_2\text{O}(\text{g}) \rightleftharpoons 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$ Given that the forward reaction (the conversion of "left-hand" species to "right-hand" species) is **endothermic**, which of the following changes will **decrease** the equilibrium amount of H_2O ?

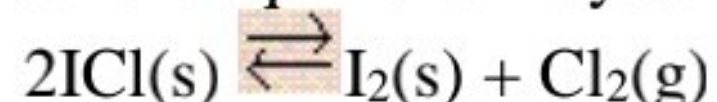
- (a) Adding more oxygen
- (b) Adding a solid phase catalyst
- (c) Decreasing the volume of the container (the total pressure increases)
- (d) Increasing the temperature at constant pressure
- (e) Adding He gas

12. The conventional equilibrium constant expression (K_c) for the system below is:

- $2\text{ICl}(\text{s}) \rightleftharpoons \text{I}_2(\text{s}) + \text{Cl}_2(\text{g})$
- (a) $[\text{I}_2][\text{Cl}_2]/[\text{ICl}]^2$
 - (b) $[\text{I}_2][\text{Cl}_2]/2[\text{ICl}]$
 - (c) $[\text{Cl}_2]$
 - (d) $([\text{I}_2] + [\text{Cl}_2])/2[\text{ICl}]$
 - (e) $[\text{Cl}_2]/[\text{ICl}]^2$

13.

Consider the equilibrium system:



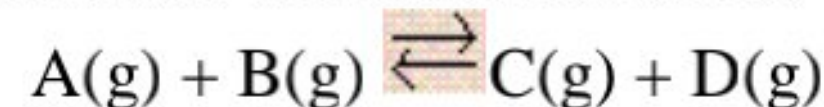
Which of the following changes will increase the total amount of Cl_2 that can be produced?

- (a) Removing some of the $\text{I}_2(\text{s})$
- (b) Adding more $\text{ICl}(\text{s})$
- (c) Removing the Cl_2 as it is formed

- (d) Decreasing the volume of the container
- (e) All of the above

14.

At equilibrium, a 1.0 liter container was found to contain 0.20 moles of A, 0.20 moles of B, 0.40 moles of C and 0.40 mole of D. If 0.10 moles of A and 0.10 moles of B are added to this system, what will be the new equilibrium concentration of A?



- (a) 0.37 mol/L
- (b) 0.47 mol/L
- (c) 0.87 mol/L
- (d) 0.23 mol/L
- (e) 0.15 mol/L

15. Consider the following system in a 1.00

L container: $\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 2\text{C}(\text{g})$ The equilibrium concentrations at 200°C were determined to be:

$$[\text{A}] = 0.200 \text{ M}$$

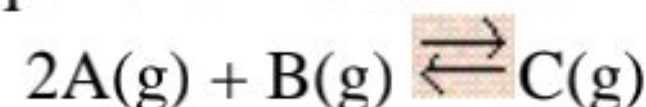
$$[\text{B}] = 3.00 \text{ M}$$

$$[\text{C}] = 0.500 \text{ M}$$

How many moles of A must be added to increase the concentration of C to 0.700 M at 200°C?

- (a) 0.225 mol
- (b) 0.305 mol
- (c) 0.417 mol
- (d) 0.610 mol
- (e) 0.700 mol

16. Consider the reversible reaction at equilibrium at 392°C:



The partial pressures are found to be: A: 6.70 atm, B: 10.1 atm, C: 3.60 atm. Evaluate K_p for this reaction.

- (a) 7.94×10^{-3}
- (b) 0.146
- (c) 0.0532
- (d) 54.5
- (e) 121

17. $K_c = 0.040$ for the system below at 450°C:



Evaluate K_p for the reaction at 450°C.

- (a) 0.40
- (b) 0.64
- (c) 2.4
- (d) 0.052
- (e) 6.7×10^{-4}

18. What is the equilibrium constant for a reaction that has a value of $\Delta G^\circ = -41.8 \text{ kJ}$ at 100°C?

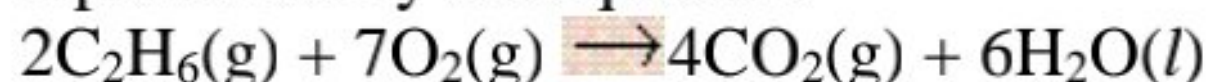
- (a) 1.01
- (b) 7.1×10^5
- (c) -5.87
- (d) 1.4×10^{-6}
- (e) 13.5

19. The equilibrium constant at 427°C for the reaction:

$$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$$
 is $K_p = 9.4 \times 10^{-5}$. Calculate the value of ΔG° for the reaction at 427°.

- (a) -33 kJ
- (b) -54 kJ

1. The combustion of ethane (C_2H_6) is represented by the equation:



In this reaction:

- (a) The rate of consumption of ethane is seven times faster than the rate of consumption of oxygen.
- (b) The rate of formation of CO_2 equals the rate of formation of water.
- (c) Water is formed at a rate equal to two-thirds the rate of formation of CO_2 .
- (d) The rate of consumption of oxygen equals the rate of consumption of water.
- (e) CO_2 is formed twice as fast as ethane is consumed.

2. The speed of a chemical reaction

- (a) is constant no matter what the temperature is.
- (b) is independent of the amount of contact surface of a solid involved.

- (c) 54 kJ
- (d) 33 kJ
- (e) 1.3 J

20. For a specific reaction, which of the following statements can be made about K , the equilibrium constant?

- (a) It always remains the same at different reaction conditions.
- (b) It increases if the concentration of one of the products is increased.
- (c) It changes with changes in the temperature.
- (d) It increases if the concentration of one of the reactants is increased.
- (e) It may be changed by the addition of a catalyst.

Answers:

- 1. (e) 2. (b) 3. (b) 4. (b) 5. (b) 6. (c) 7. (a) 8. (c) 9. (a) 10. (c) 11. (d) 12. (c) 13. (c) 14. (d) 15. (b) 16. (a) 17. (c) 18. (b) 19. (c) 20. (c)**

(c) Between gases should in all cases be extremely rapid because the average kinetic energy of the molecules is great.

(d) Between ions in aqueous solution is extremely rapid because there are no bonds that need to be broken.

(e) Varies inversely with the absolute temperature.

3. For a reaction $2\text{A} + \text{B} \rightarrow 2\text{C}$, with the rate equation: $\text{Rate} = k[\text{A}]^2[\text{B}]$

- (a) The order with respect to A is 1 and the order overall is 1.
- (b) The order with respect to A is 2 and the order overall is 2.
- (c) The order with respect to A is 2 and the order overall is 3.
- (d) The order with respect to B is 2 and the order overall is 2.
- (e) The order with respect to B is 2 and the order overall is 3.

4. Given the following data for this reaction:
 $\text{NH}_4^+(\text{aq}) + \text{NO}_2^-(\text{aq}) \rightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$

EXPT	[NH ₄ ⁺]	[NO ₂ ⁻]	RATE
1	0.010 M	0.020 M	0.020 M/s
2	0.015 M	0.020 M	0.030 M/s
3	0.010 M	0.010 M	0.005 M/s

The rate law for the reaction is:

- (a) Rate = $k [\text{NH}_4^+] [\text{NO}_2^-]$
- (b) Rate = $k [\text{NH}_4^+]^2 [\text{NO}_2^-]^2$
- (c) Rate = $k [\text{NH}_4^+]^2 [\text{NO}_2^-]$
- (d) Rate = $k [\text{NH}_4^+] [\text{NO}_2^-]^2$
- (e) None of the above

5. What are the units of k for the rate law: Rate = $k[\text{A}][\text{B}]^2$, when the concentration unit is mol/L?

- (a) s⁻¹
- (b) s
- (c) L mol⁻¹ s⁻¹
- (d) L² mol⁻² s⁻¹
- (e) L² s² mol⁻²

6. Given: $\text{A} + 3\text{B} \rightarrow 2\text{C} + \text{D}$

This reaction is first order with respect to reactant A and second order with respect to reactant B. If the concentration of A is doubled and the concentration of B is halved, the rate of the reaction would _____ by a factor of _____.

- (a) Increase, 2
- (b) Decrease, 2
- (c) Increase, 4
- (d) Decrease, 4
- (e) Not change

7. The decomposition of carbon disulfide, CS₂, to carbon monosulfide, CS, and sulfur is first order with $k = 2.8 \times 10^{-7} \text{ s}^{-1}$ at 1000°C. $\text{CS}_2 \rightarrow \text{CS} + \text{S}$

What is the half-life of this reaction at 1000°C?

- (a) $5.0 \times 10^7 \text{ s}$
- (b) $4.7 \times 10^{-6} \text{ s}$
- (c) $3.8 \times 10^5 \text{ s}$
- (d) $6.1 \times 10^4 \text{ s}$
- (e) $2.5 \times 10^6 \text{ s}$

8. The decomposition of dimethylether at 504 °C is first order with a half-life of 1570 seconds. What fraction of an initial amount

of dimethylether remains after 4710 seconds?

- (a) 1/3
- (b) 1/6
- (c) 1/8
- (d) 1/16
- (e) 1/32

9. The half-life for a first-order reaction is 32 s. What was the original concentration if, after 2.0 minutes, the reactant concentration is 0.062 M?

- (a) 0.84 M
- (b) 0.069 M
- (c) 0.091 M
- (d) 0.075 M
- (e) 0.13 M

10. Given that a reaction absorbs energy and has an activation energy of 50 kJ/mol, which of the following statements are **correct**? (Hint: Draw the potential energy diagram.)

- (1) The reverse reaction has an activation energy equal to 50 kJ/mol.
- (2) The reverse reaction has an activation energy less than 50 kJ/mol.
- (3) The reverse reaction has an activation energy greater than 50 kJ/mol.
- (4) The change in internal energy is less than zero.
- (5) The change in internal energy is greater than zero.

- (a) (1) and (4)
- (b) (2) and (4)
- (c) (3) and (4)
- (d) (2) and (5)
- (e) (3) and (5)

11. If reaction A has an activation energy of 250 kJ and reaction B has an activation energy of 100 kJ, which of the following statements **must** be correct?

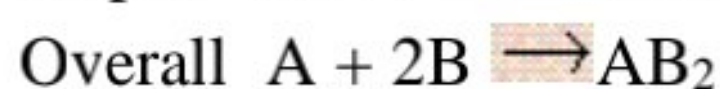
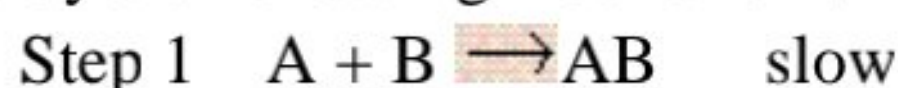
- (a) If reaction A is exothermic and reaction B is endothermic then reaction A is favored kinetically.
- (b) At the same temperature the rate of reaction B is greater than the rate of reaction A.

- (c) The energy of reaction A must be greater than the energy of reaction B.
 (d) The energy of reaction B must be greater than the energy of reaction A.
 (e) The rate of reaction A at 25 °C equals the rate of reaction B at 100 °C.

12. If the activation energy in the forward direction of an elementary step is 52 kJ and the activation energy in the reverse direction is 74 kJ, what is the energy of reaction ΔE for this step?

- (a) 22 kJ
 (b) -22 kJ
 (c) 52 kJ
 (d) -52 kJ
 (e) 126 kJ

13. Suppose the reaction: $A + 2B \rightarrow AB_2$ occurs by the following mechanism:



The rate law expression must be Rate =

- (a) $k[A]$
 (b) $k[B]$
 (c) $k[A][B]$
 (d) $k[B]^2$
 (e) $k[A][B]^2$

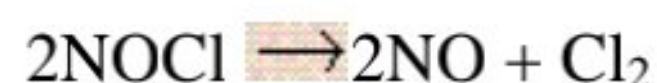
14. A possible mechanism for the reaction, $2A + B \rightarrow C + D$, is:

- (1) $A + A \rightleftharpoons A_2$ fast, equilibrium
 (2) $A_2 + A \rightarrow A_3$ slow
 (3) $A_3 + B \rightarrow A + C + D$ fast

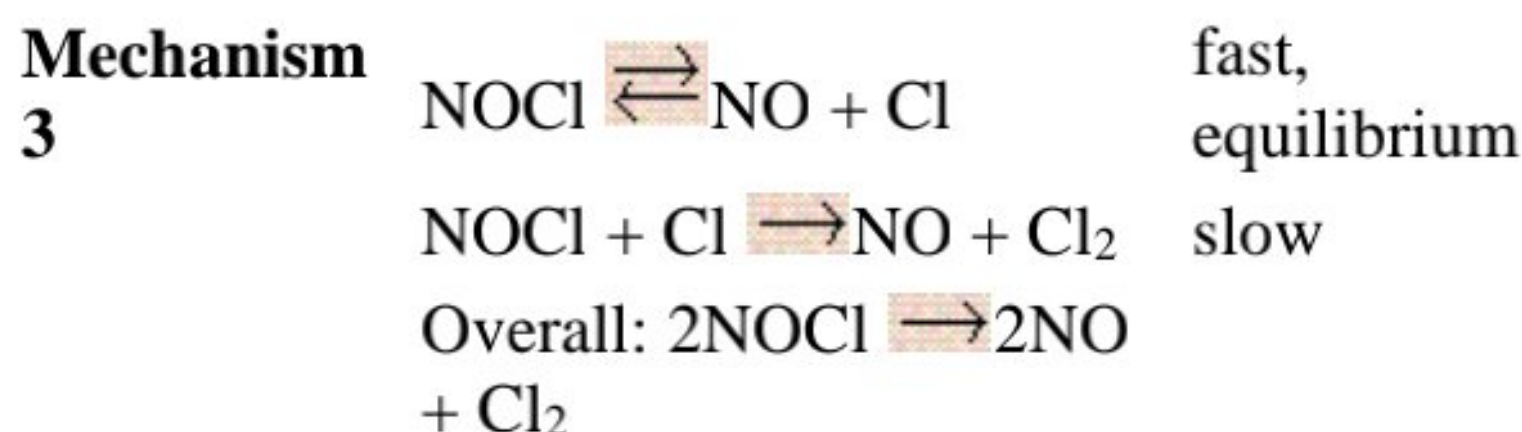
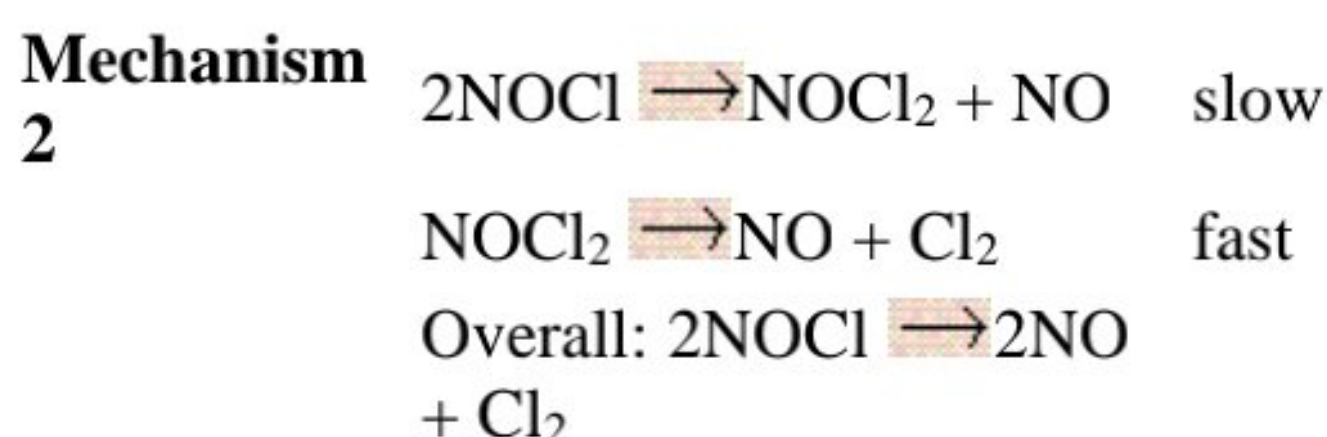
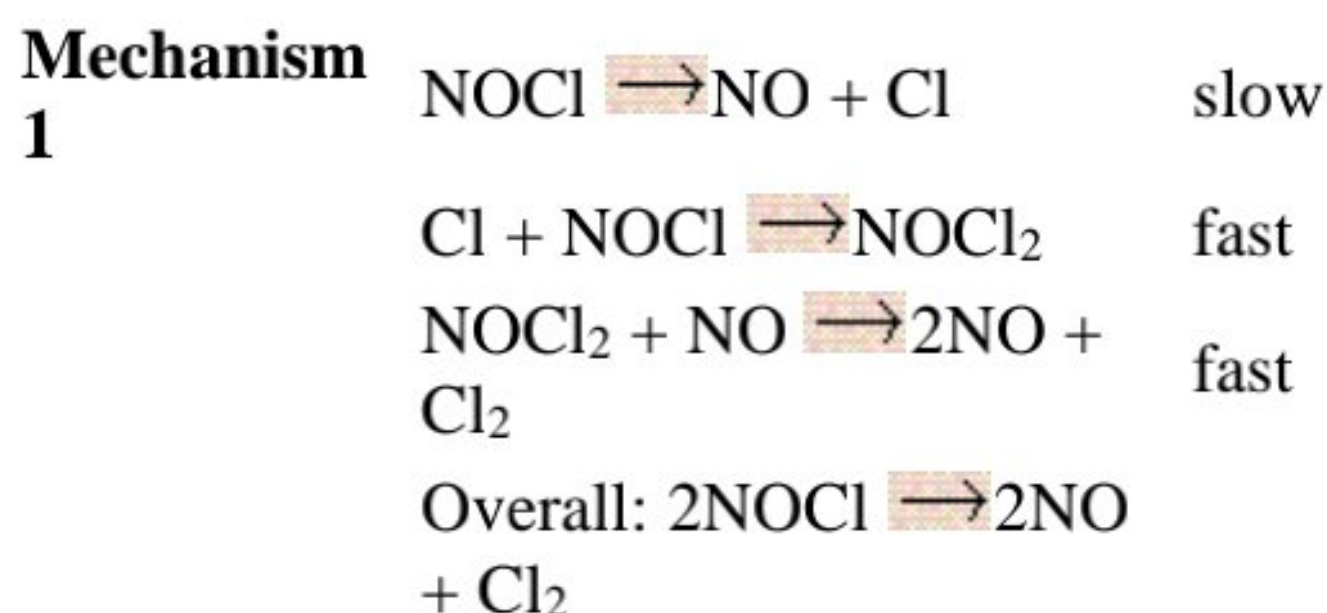
According to the mechanism, the rate law will be:

- (a) Rate = $k[A]^2$
 (b) Rate = $k[A][B]$
 (c) Rate = $k[A]^2[B]$
 (d) Rate = $k[A]$
 (e) Rate = $k[A]^3$

15. At 300 K, the following reaction is found to obey the rate law: Rate = $k[NOCl]^2$:



Consider the three postulated mechanisms given below. Then choose the response that lists all those that are possibly correct and no others.



- (a) 2, 3
 (b) 3
 (c) 1
 (d) 2
 (e) 1, 2

16. A correct reaction mechanism for a given reaction usually is:

- (a) The same as its balanced chemical equation.
 (b) Obvious if its heat of reaction is known.
 (c) Obvious if its reaction order is known.
 (d) Sometimes difficult to prove.
 (e) Obvious if its activation energy is known.

17. Suppose the activation energy of a certain reaction is 250 kJ/mol. If the rate constant at $T_1 = 300$ K is k_1 and the rate constant at $T_2 = 320$ K is k_2 , then the reaction is ___ times faster at 320 K than at 300 K. (Hint: Solve for k_2/k_1 .)

- (a) 3×10^{-29}
 (b) 0.067

- (c) 15.0
- (d) 525
- (e) 3×10^{-28}

18. What is the activation energy (in kJ) of a reaction whose rate constant increases by a factor of 100 upon increasing the temperature from 300 K to 360 K?

- (a) 27
- (b) 35
- (c) 42
- (d) 53
- (e) 69

19. Most reactions are more rapid at high temperatures than at low temperatures. This is **consistent** with: (I) an increase in the activation energy with increasing temperature.

(II) an increase in the rate constant with increasing temperatures.

(III) An increase in the percentage of "high energy" collisions with increasing temperature.

- (a) Only I
- (b) Only II
- (c) Only III
- (d) Only I and II
- (e) Only II and III

20. Which items correctly complete the following statement?

A catalyst can act in a chemical reaction to:

(I) increase the equilibrium constant. (II) Lower the activation energy. (III) Decrease ΔE for the reaction. (IV) Provide a new path for the reaction.

- (a) Only I & II
- (b) Only II & III
- (c) Only III & IV
- (d) Only I & III
- (e) Only II & IV

21. A catalyst:

- (a) Actually participates in the reaction.
- (b) Changes the equilibrium concentration of the products.
- (c) Does not affect a reaction energy path.
- (d) Always decreases the rate for a reaction.

- (e) Always increases the activation energy for a reaction.

22. Which statement is **false**?

- (a) If a reaction is thermodynamically spontaneous it may occur rapidly.
- (b) If a reaction is thermodynamically spontaneous it may occur slowly.
- (c) Activation energy is a kinetic quantity rather than a thermodynamic quantity.
- (d) If a reaction is thermodynamically nonspontaneous, it will not occur spontaneously.
- (e) If a reaction is thermodynamically spontaneous, it must have a low activation energy.

23. Which of the following statements are **true**?

- (1) Reactions with more negative values of ΔG° are spontaneous and proceed at a higher rate than those with less negative values of ΔG° .
- (2) The activation energy, E_a , is usually about the same as ΔE for a reaction.
- (3) The activation energy for a reaction does not change significantly as temperature changes.
- (4) Reactions usually occur at faster rates at higher temperatures.

- (a) 1, 2, 4
- (b) 3, 4
- (c) 1, 2, 3
- (d) 2, 3, 4
- (e) 1, 2, 3, 4

24. When the concentration of reactant molecules is increased, the rate of reaction increases. The best explanation is: As the reactant concentration increases, (a) the average kinetic energy of molecules increases.

- (b) The frequency of molecular collisions increases.
- (c) The rate constant increases.
- (d) The activation energy increases.
- (e) The order of reaction increases.

25. For the reaction, $2\text{H}_2\text{S}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{S}(\text{s}) + 2\text{H}_2\text{O}(\text{l})$, which one of the following statements is **absolutely** true?

- (a) The reaction is first order with respect to H_2S and second order with respect to O_2 .
- (b) The reaction is fourth order overall.
- (c) The rate law is: $\text{rate} = k[\text{H}_2\text{S}]^2[\text{O}_2]$.
- (d) The rate law is: $\text{rate} = k[\text{H}_2\text{S}][\text{O}_2]$.

(e) The rate law cannot be determined from the information given.

Answers:

1. (e) 2. (d) 3. (c) 4. (d) 5. (d) 6. (b) 7. (e) 8. (c) 9. (a) 10. (d) 11. (b) 12. (b) 13. (c) 14. (e) 15. (d) 16. (d) 17. (d) 18. (e) 19. (e) 20. (e) 21. (a) 22. (e) 23. (b) 24. (b) 25. (e)

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